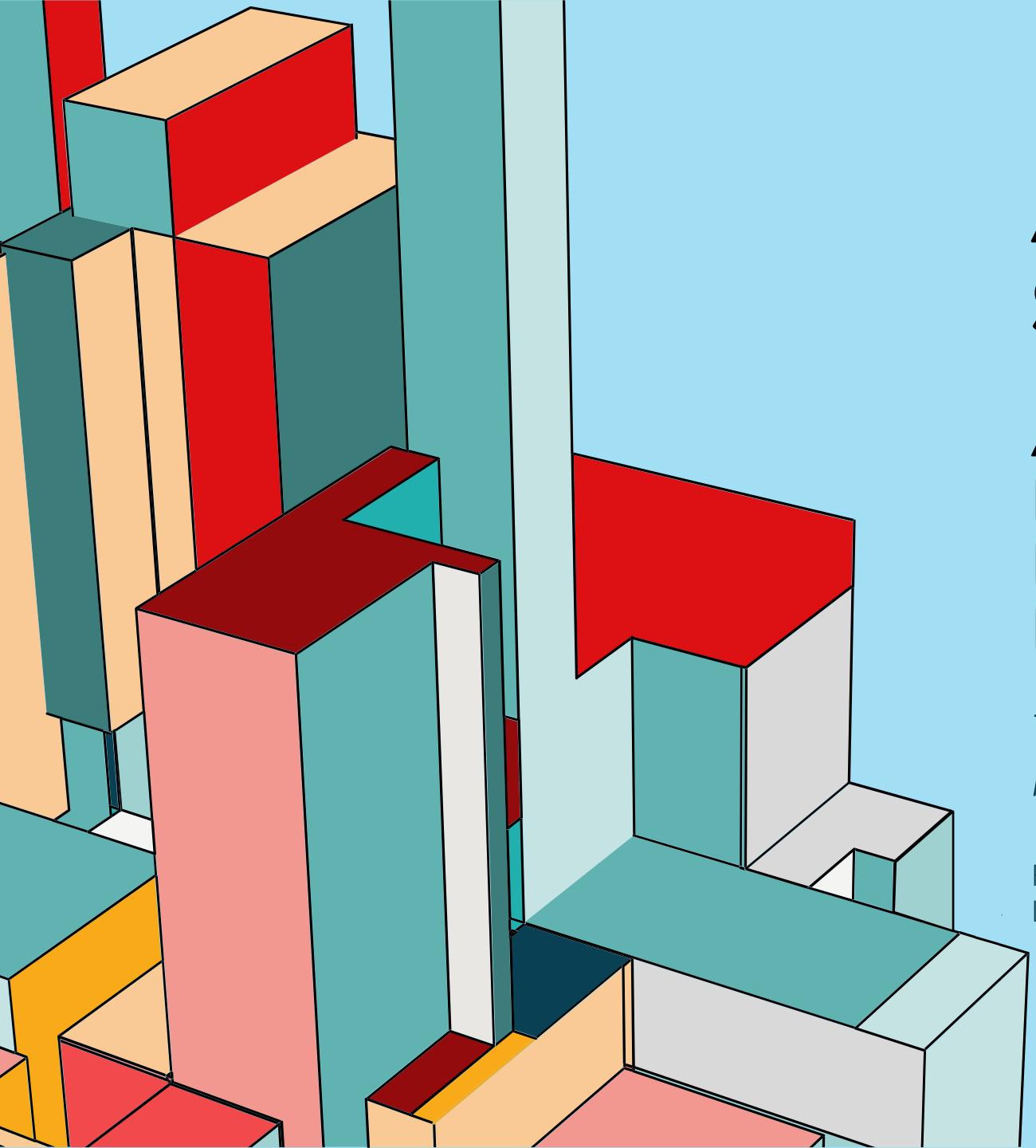


APPLIED DATA SCIENCE PROJECT

ADAPTING MANIPULATION DATASETS TO A HUMANOID ROBOT

Teach a humanoid new tricks with another robot's demonstrations

Proposed by **Lorenzo Natale**
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FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

Humanoids are data-hungry, let's recycle robot data

Why

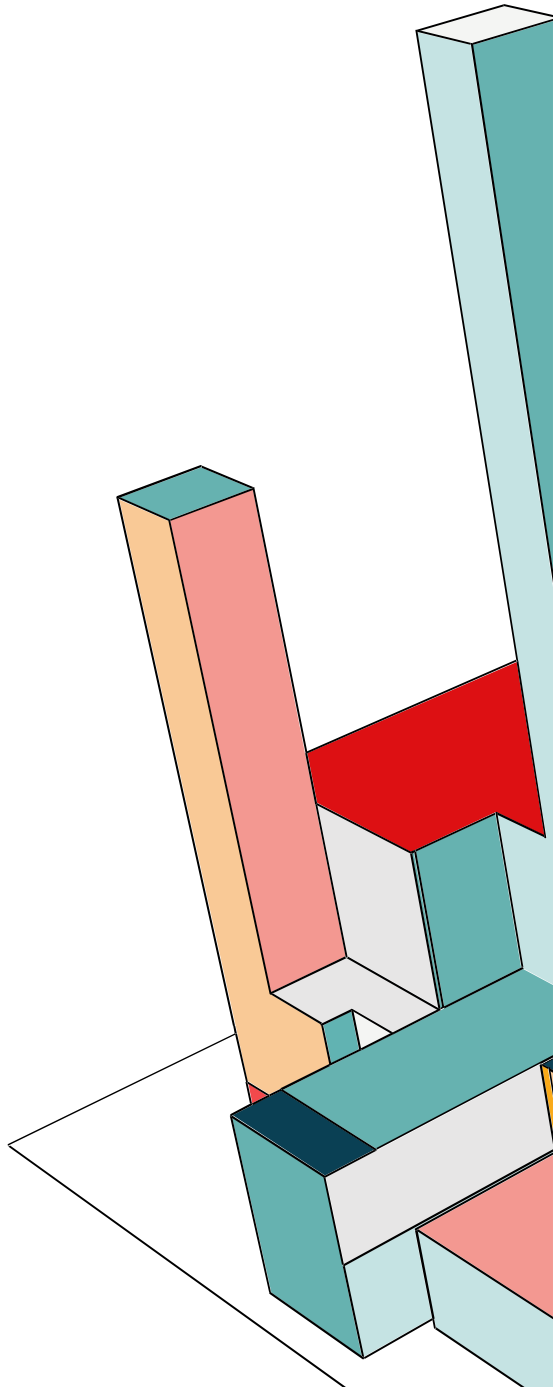
Big manipulation datasets come from other robots. Humanoids have little data and are costly to teleoperate, so reusing demonstrations cuts training costs.

Who benefits

Workers in physically demanding jobs, people who need help at home or in care, and companies that could adopt humanoids sooner and at lower cost.

UN SDGs

SDG 9: Industry, Innovation and Infrastructure.
Indirectly SDG 8: Decent Work, as humanoids take on physically demanding tasks.



DATA

From one robot's demonstrations to a humanoid-ready dataset

Source

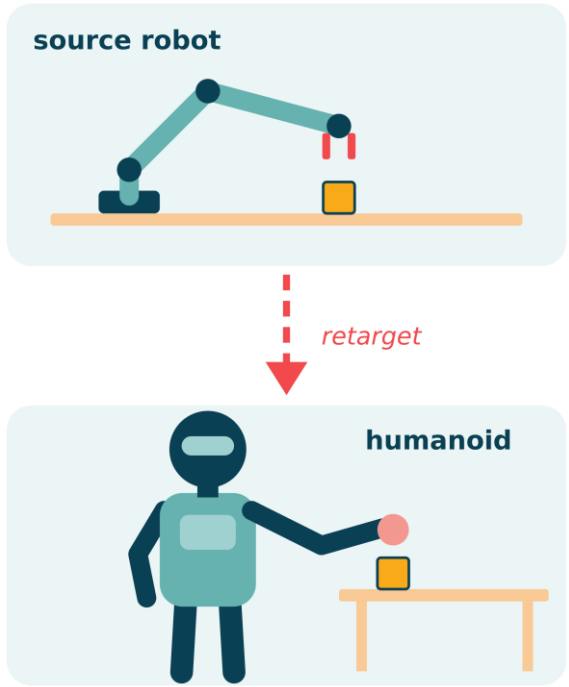
Demonstrations from a simulated benchmark, from LIBERO-10 tasks (50 teleoperated demos each, recorded on a Franka arm) to Isaac Sim scenarios.

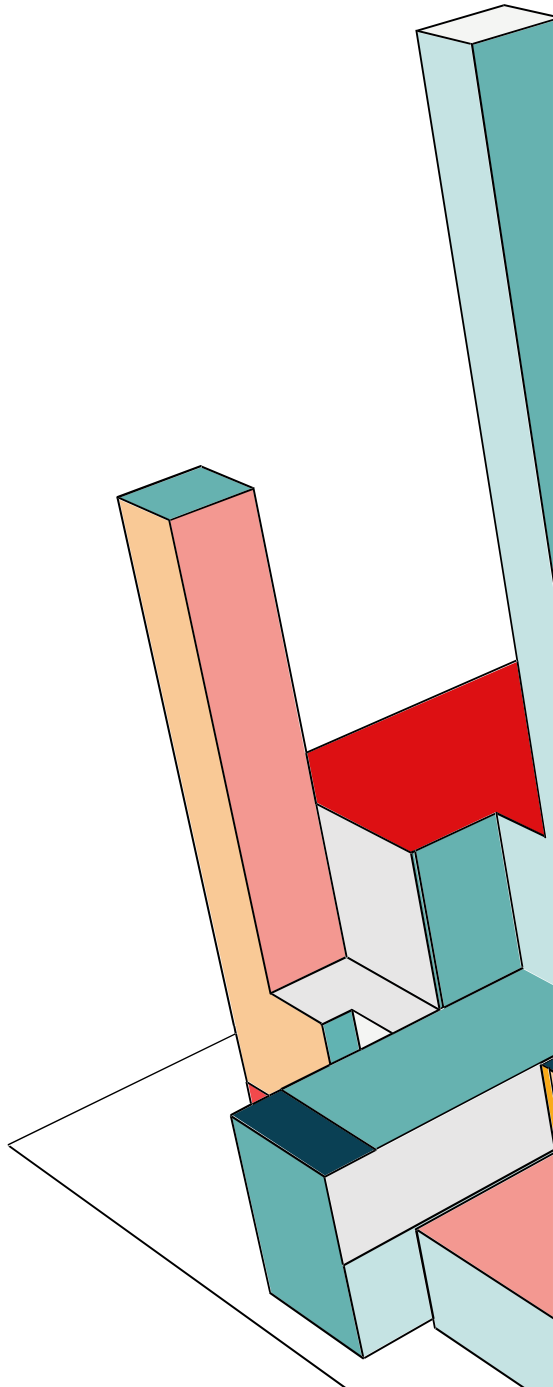
Build

Retarget trajectories to ergoCub or R1 (model provided by IIT) through inverse kinematics, map grippers to hands, and re-render the humanoid's camera views.

Check

Replay every adapted demo in simulation: joint limits, reachability, collisions, task completion. Keep the feasible ones and report how many survive.





TASK

Retarget, verify, learn, measure



ML task

Imitation learning, with a policy ranging from ACT to GR00T N1.7, trained from scratch or fine-tuned.

Metrics

Task success rate over repeated randomized trials · share of demos still feasible after retargeting · end-effector tracking error vs. the source.

Benchmark and policy chosen according to available compute, compatible checkpoints and training data.

